

VEDA TECHNOLOGY

Data Analytics Internship Programme

Product Basket Analysis

Finding products that are frequently purchased together

Task 29 | Level 2 | Day 29 of 45

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Tools used

Python | Pandas | Matplotlib | Google Colab

1. Introduction

Retail businesses collect thousands of transactions every day, and hidden inside this data are useful buying patterns. Product basket analysis (also called market basket analysis) studies which products customers tend to buy **together** in the same order. Such patterns help a business recommend related products, create bundle offers and place items sensibly on its website or in its store.

Objective

The objective of this task is to introduce association-style analysis. Using order-level data, the task identifies the product pairs that are bought together most often and measures the strength of each relationship, so that simple product recommendations can be produced.

Deliverables

- **Top product pairs** that are frequently purchased together.
- **Recommendations** built from the strongest associations.

Dataset and Tools

The analysis uses an Online Retail II style transactional dataset with the columns Invoice, StockCode, Description, Quantity, InvoiceDate, Price, Customer ID and Country. The file loaded in the notebook contains **5,482 rows** and **1,500 unique invoices**. The work was done in Google Colab using Python, with Pandas for data handling, the collections and itertools modules for counting pairs, and Matplotlib for charts.

Item	Detail
Dataset	Online Retail II style transaction data
Rows loaded	5,482
Rows after cleaning	5,482
Products analysed	200 most frequently sold products
Orders with 2 or more products	1,327
Environment	Google Colab (Python 3)

2. Methodology

The analysis followed five clear steps. Each step matches a block of code in the Colab notebook.

Step 1: Load data

The CSV file was read into a Pandas DataFrame and its shape and column names were checked.

Step 2: Clean data

Rows with missing descriptions were dropped. Cancelled orders (invoice numbers starting with C) and rows with zero or negative quantity or price were removed. Non-product codes such as POST, DOT, M and BANK CHARGES were excluded.

Step 3: Build baskets

The 200 most popular products were kept for speed. Transactions were grouped by **invoice number**, giving one list of unique products per order. Orders with fewer than two products were dropped.

Step 4: Count pairs

For every order, all unique product pairs were generated with `combinations(items, 2)`, which automatically **excludes self-pairs**. A Counter then counted how often each pair appeared.

Step 5: Measure strength

Support, Confidence and Lift were calculated for every pair that appeared in at least 50 orders, in both directions (A to B and B to A).

Metrics used

Metric	Meaning	Formula
Support	Share of all orders that contain both products	$\text{count}(A,B) / \text{total orders}$
Confidence	Of the orders containing A, the share that also contain B	$\text{count}(A,B) / \text{count}(A)$
Lift	How much more often A and B are bought together than by chance. A value above 1 means a real association	$\text{Confidence} / (\text{count}(B) / \text{total orders})$

3. Results: Top Product Pairs

After cleaning and filtering, **1,327 orders** contained at least two products. The most frequently purchased pair was **Paper Chain Kit 50'S Christmas** and **Party Bunting**, bought together in **137 orders**.

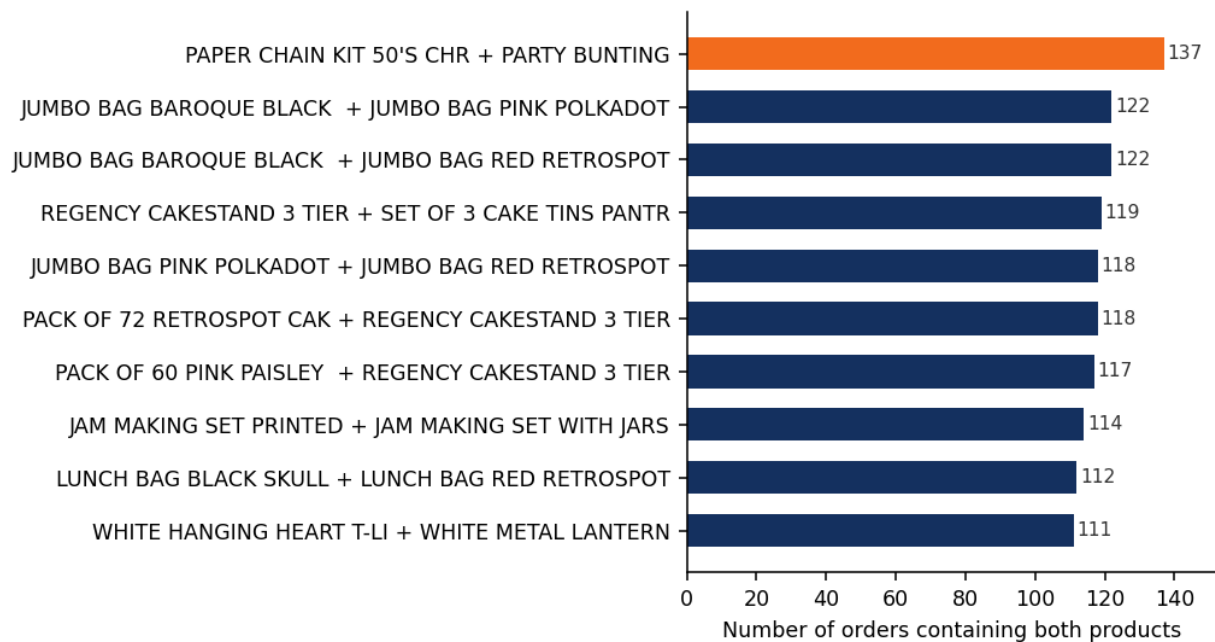


Figure 1: Ten most frequently purchased product pairs

#	Product 1	Product 2	Orders
1	PAPER CHAIN KIT 50'S CHRISTMAS	PARTY BUNTING	137
2	JUMBO BAG BAROQUE BLACK WHITE	JUMBO BAG PINK POLKADOT	122
3	JUMBO BAG BAROQUE BLACK WHITE	JUMBO BAG RED RETROSPOT	122
4	REGENCY CAKESTAND 3 TIER	SET OF 3 CAKE TINS PANTRY DESIGN	119
5	JUMBO BAG PINK POLKADOT	JUMBO BAG RED RETROSPOT	118
6	PACK OF 72 RETROSPOT CAKE CASES	REGENCY CAKESTAND 3 TIER	118
7	PACK OF 60 PINK PAISLEY CAKE CASES	REGENCY CAKESTAND 3 TIER	117
8	JAM MAKING SET PRINTED	JAM MAKING SET WITH JARS	114
9	LUNCH BAG BLACK SKULL	LUNCH BAG RED RETROSPOT	112
10	WHITE HANGING HEART T-LIGHT HOLDER	WHITE METAL LANTERN	111
11	CREAM CUPID HEARTS COAT HANGER	WHITE HANGING HEART T-LIGHT HOLDER	109
12	PACK OF 60 PINK PAISLEY CAKE CASES	PACK OF 72 RETROSPOT CAKE CASES	109
13	LUNCH BAG RED RETROSPOT	LUNCH BAG SPACEBOY DESIGN	109
14	LUNCH BAG BLACK SKULL	LUNCH BAG SPACEBOY DESIGN	107
15	GLASS STAR FROSTED T-LIGHT HOLDER	WHITE HANGING HEART T-LIGHT HOLDER	103

4. Results: Association Strength

Frequency alone can be misleading because very popular products appear together simply by chance. Lift corrects for this. All top pairs below have a Lift well above 1, which confirms that these products are genuinely related.

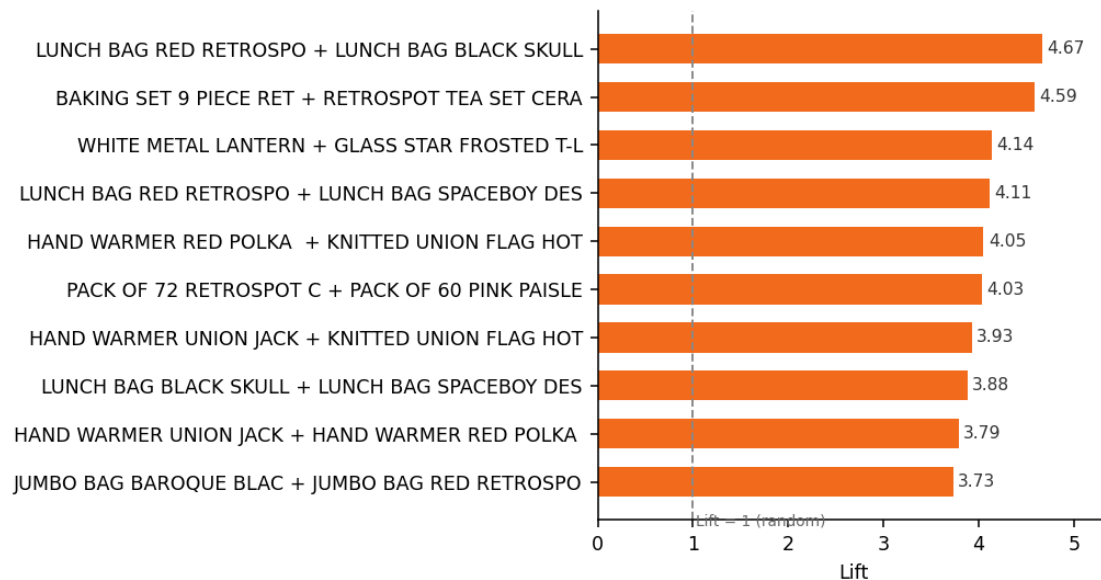


Figure 2: Ten strongest associations ranked by Lift

Product A	Product B	Count	Support	Confidence	Lift
LUNCH BAG RED RETROSPOT	LUNCH BAG BLACK SKULL	112	0.0844	0.640	4.67
LUNCH BAG BLACK SKULL	LUNCH BAG RED RETROSPOT	112	0.0844	0.615	4.67
BAKING SET 9 PIECE RETROSPOT	RETROSPOT TEA SET CERAMIC 11 PC	98	0.0739	0.563	4.59
RETROSPOT TEA SET CERAMIC 11 PC	BAKING SET 9 PIECE RETROSPOT	98	0.0739	0.601	4.59
WHITE METAL LANTERN	GLASS STAR FROSTED T-LIGHT HOLDER	101	0.0761	0.577	4.14
GLASS STAR FROSTED T-LIGHT HOLDER	WHITE METAL LANTERN	101	0.0761	0.546	4.14
LUNCH BAG RED RETROSPOT	LUNCH BAG SPACEBOY DESIGN	109	0.0821	0.623	4.11
LUNCH BAG SPACEBOY DESIGN	LUNCH BAG RED RETROSPOT	109	0.0821	0.542	4.11
HAND WARMER RED POLKA DOT	KNITTED UNION FLAG HOT WATER BOTTLE	95	0.0716	0.537	4.05
KNITTED UNION FLAG HOT WATER BOTTLE	HAND WARMER RED POLKA DOT	95	0.0716	0.540	4.05

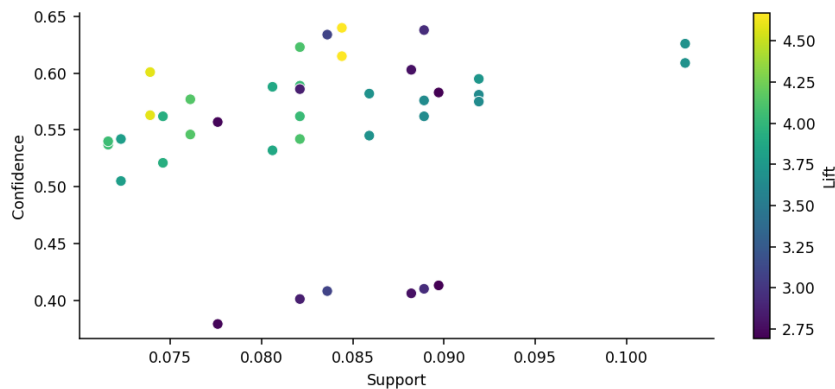


Figure 3: Support versus Confidence for all rules (colour shows Lift)

5. Recommendations

A small function, `recommend(product)`, returns the products most likely to be bought together with a chosen item, ranked by Confidence. Some examples:

If the customer buys	Recommend	Confidence	Lift
PAPER CHAIN KIT 50'S CHRISTMAS	PARTY BUNTING	0.626	3.69
LUNCH BAG RED RETROSPOT	LUNCH BAG BLACK SKULL	0.640	4.67
LUNCH BAG RED RETROSPOT	LUNCH BAG SPACEBOY DESIGN	0.623	4.11
REGENCY CAKESTAND 3 TIER	SET OF 3 CAKE TINS PANTRY DESIGN	0.413	2.69
REGENCY CAKESTAND 3 TIER	PACK OF 72 RETROSPOT CAKE CASES	0.410	2.94
WHITE METAL LANTERN	WHITE HANGING HEART T-LIGHT HOLDER	0.634	3.09
WHITE METAL LANTERN	GLASS STAR FROSTED T-LIGHT HOLDER	0.577	4.14

Business use

- **Customers also bought:** show the recommended item on the product and checkout pages.
- **Bundle offers:** package strongly linked items, such as matching lunch bags or the baking set with the tea set, at a small discount.
- **Placement:** keep related items close together in the store or on the same web page.
- **Seasonal campaigns:** promote items like paper chains and party bunting together before festive periods.

6. Conclusion

This task applied association-style analysis to order data. After cleaning, 1,327 multi-product orders were analysed. The most frequent pair was Paper Chain Kit 50'S Christmas with Party Bunting (137 orders). The strongest association was between Lunch Bag Red Retrospot and Lunch Bag Black Skull, with a Lift of 4.67, meaning they are bought together about 4.7 times more often than by chance.

Most strong pairs belong to the same product family or design theme, which shows that customers prefer to buy matching or complementary items together. These findings can directly support recommendations, bundles and cross-selling to raise the average order value.

Limitations and next steps: only pairs were studied, and only the top 200 products were included. A future version could use the Apriori algorithm to find larger item sets (three or more products) and could include the full product range.